

BÀI TẬP VỀ NHÀ

19/05/2026

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Bài 1: file_server

- Chạy file_server, dùng netcat client để kết nối server nhận được số file và tên các file đang có tại server
 - o Nếu server có file

The screenshot displays the Visual Studio Code interface with a C++ file server implementation. The Explorer sidebar on the left shows a project structure with folders for 'file_server' and 'file_server.c', and a file named 'file_server.c'. The main editor area shows the source code of 'file_server.c', which includes a header file, a main function, and a loop that reads files from the server and sends them to the client. The terminal at the bottom shows the execution of the program, which outputs 'Server is listening on port 9000...' and 'New client connected: 5'. The output also shows the client sending a request for 'hello_world.txt' and receiving the file content.

```
char header[64];
int header_len = sprintf(header, "OK %d\r\n", file_size);
send(client, header, header_len, 0);

char file_buf[1024];
int bytes_read;

// read file and send to client
while ((bytes_read = fread(file_buf, 1, sizeof(file_buf), f)) > 0)
{
    int bytes_sent = 0;
    while (bytes_sent < bytes_read)
    {
        int ret = send(client, file_buf + bytes_sent, bytes_read - bytes_sent, 0);
        bytes_sent += ret;
    }
}

// close file
fclose(f);
}
```

Terminal Output:

```
nc -l -p 9000
Server is listening on port 9000...
New client connected: 5
OK 2
hello_world.txt
test.txt
```

- o Nếu server không có file

```
File Edit Selection View Go Run Terminal Help
C:\file_server V X hello_world V
bt-1205 > C:\file_server> main()
52 {
53     perror("listen() failed");
54     close(listener);
55     exit(1);
56 }
57
58 printf("Server is listening on port 9000...\n");
59
60 // open folder
61 DIR *d;
62 struct dirent *dir;
63 d = opendir("./file");
64
65 // handler close process
66 signal(SIGCHLD, signalHandler);
67 while (1)
68 {
69     int client = accept(listener, NULL, NULL);
70     if (fork() == 0) // on child process
71     {
72         close(listener);
73
74         printf("New client connected: %d\n", client);
75
76         char buf[1024];
77         int buf_size = 0;
78     }
79 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

newai@DESKTOP-DEBWPSP:/mnt/d/code-documents/C++/itm-20252/bt-1205\$ cd "/mnt/d/code-documents/C++/itm-20252/bt-1205/" && gcc file_server.c -o file_server && "/mnt/d/code-documents/C++/itm-20252/bt-1205/file_server"

Server is listening on port 9000...

New client connected: 5

Child process 4140 terminated.

newai@DESKTOP-DEBWPSP:/mnt/d/code-documents/C++/itm-20252\$ nc 127.0.0.1 9000

example.txt

No files to download

- Client gửi tên file muốn tải
 - o File không tồn tại

```
File Edit Selection View Go Run Terminal Help
C:\file_server V X hello_world V
bt-1205 > C:\file_server> main()
52 {
53     perror("listen() failed");
54     close(listener);
55     exit(1);
56 }
57
58 printf("Server is listening on port 9000...\n");
59
60 // open folder
61 DIR *d;
62 struct dirent *dir;
63 d = opendir("./files");
64
65 // handler close process
66 signal(SIGCHLD, signalHandler);
67 while (1)
68 {
69     int client = accept(listener, NULL, NULL);
70     if (fork() == 0) // on child process
71     {
72         close(listener);
73
74         printf("New client connected: %d\n", client);
75
76         char buf[1024];
77         int buf_size = 0;
78     }
79 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

newai@DESKTOP-DEBWPSP:/mnt/d/code-documents/C++/itm-20252/bt-1205\$ cd "/mnt/d/code-documents/C++/itm-20252/bt-1205/" && gcc file_server.c -o file_server && "/mnt/d/code-documents/C++/itm-20252/bt-1205/file_server"

Server is listening on port 9000...

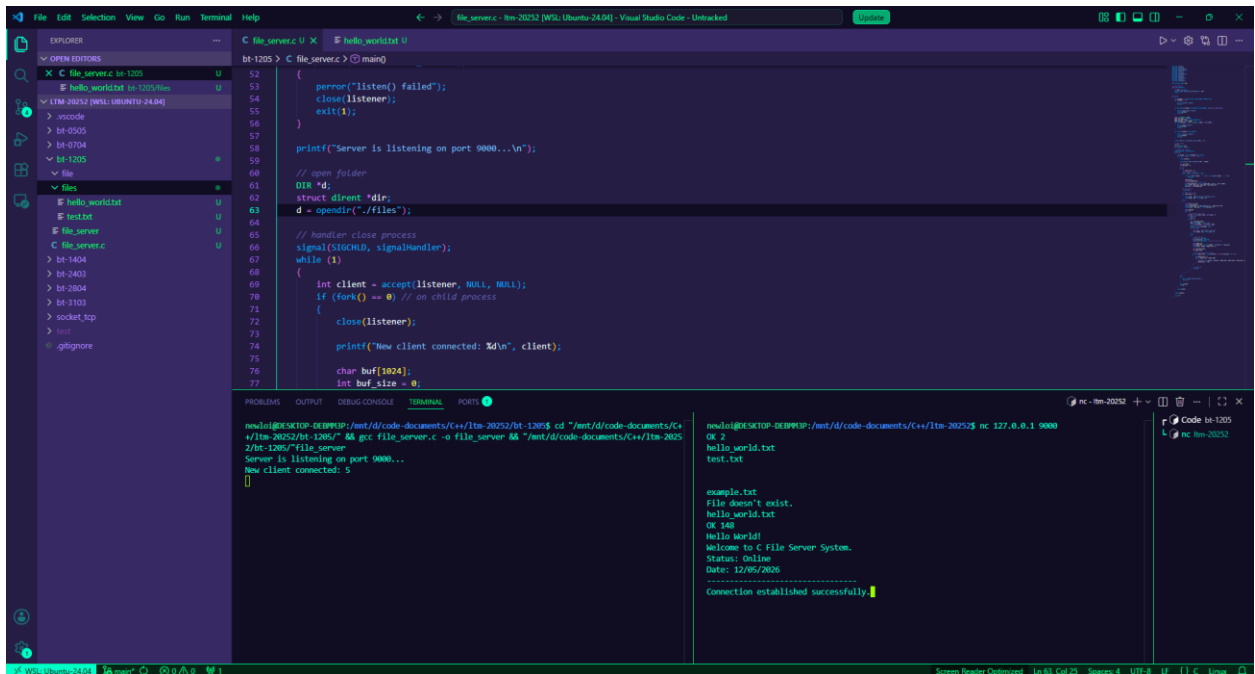
New client connected: 5

newai@DESKTOP-DEBWPSP:/mnt/d/code-documents/C++/itm-20252\$ nc 127.0.0.1 9000

example.txt

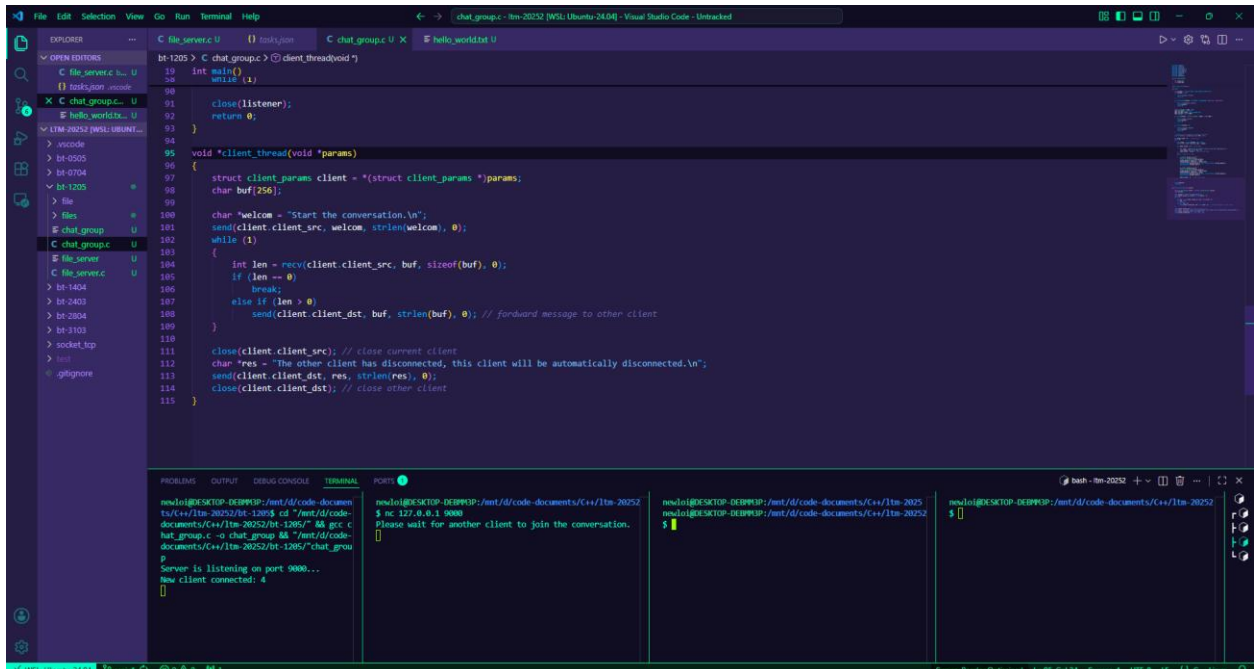
File doesn't exist.

- o File tồn tại, nhận được nội dung file



Bài 2: chat_group

- Chạy `chat_group` server và kết nối 1 client, sẽ nhận được thông báo đợi thêm 1 client nữa để bắt đầu chat



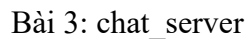
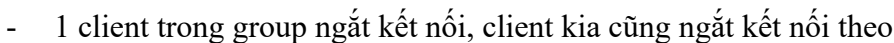
- Kết nối thêm 1 client để bắt đầu chat, 2 client sẽ nhận được thông báo bắt đầu chat

```
bt-1205 > C chat_group > client_threadvoid *)
10 int main()
11 while (1)
12
13 close(listener);
14 return 0;
15
16 void *client_thread(void *params)
17 {
18     struct client_params client = *(struct client_params *)params;
19     char buf[256];
20
21     char *welcome = "Start the conversation.\n";
22     send(client.client_src, welcome, strlen(welcome), 0);
23     while (1)
24     {
25         int len = recv(client.client_src, buf, sizeof(buf), 0);
26         if (len == 0)
27             break;
28         else if (len > 0)
29             send(client.client_dst, buf, strlen(buf), 0); // forward message to other client
30     }
31     close(client.client_src); // close current client
32     char *res = "The other client has disconnected, this client will be automatically disconnected.\n";
33     send(client.client_dst, res, strlen(res), 0);
34     close(client.client_dst); // close other client
35 }
```

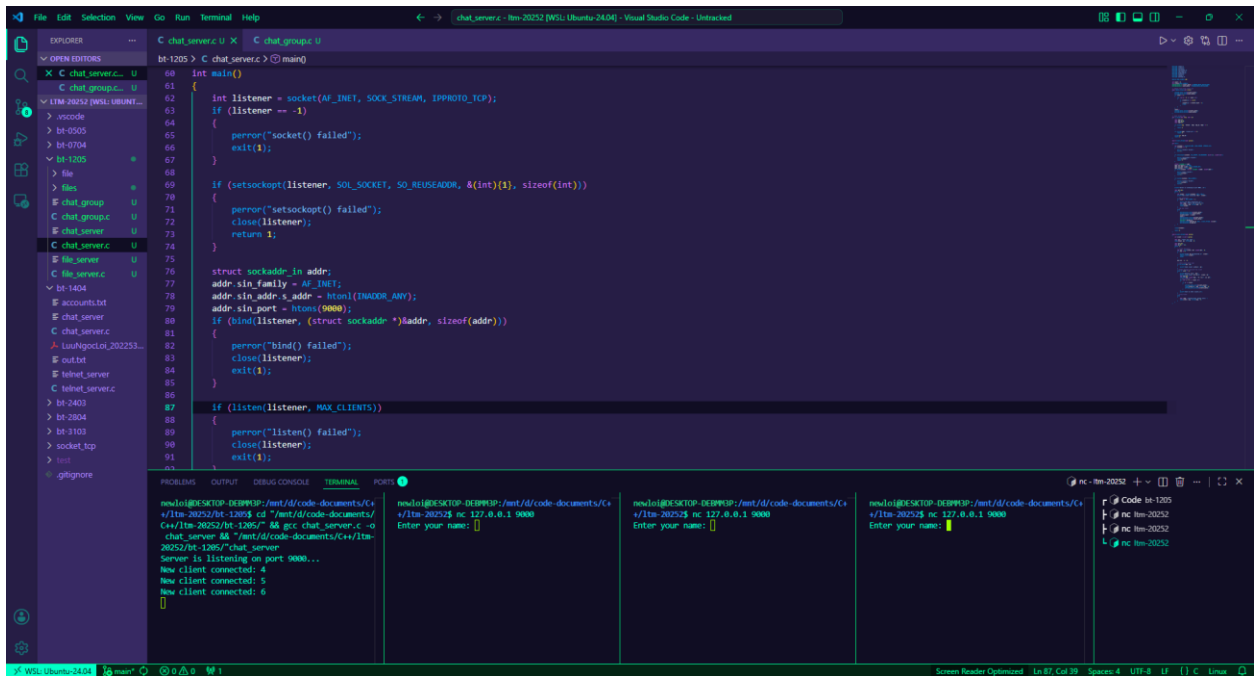
- Kết nối thêm 1 client nữa, client này chưa được ghép cặp với client nào nên không nhận được tin nhắn

```
bt-1205 > C chat_group > client_threadvoid *)
10 int main()
11 while (1)
12
13 close(listener);
14 return 0;
15
16 void *client_thread(void *params)
17 {
18     struct client_params client = *(struct client_params *)params;
19     char buf[256];
20
21     char *welcome = "Start the conversation.\n";
22     send(client.client_src, welcome, strlen(welcome), 0);
23     while (1)
24     {
25         int len = recv(client.client_src, buf, sizeof(buf), 0);
26         if (len == 0)
27             break;
28         else if (len > 0)
29             send(client.client_dst, buf, strlen(buf), 0); // forward message to other client
30     }
31     close(client.client_src); // close current client
32     char *res = "The other client has disconnected, this client will be automatically disconnected.\n";
33     send(client.client_dst, res, strlen(res), 0);
34     close(client.client_dst); // close other client
35 }
```

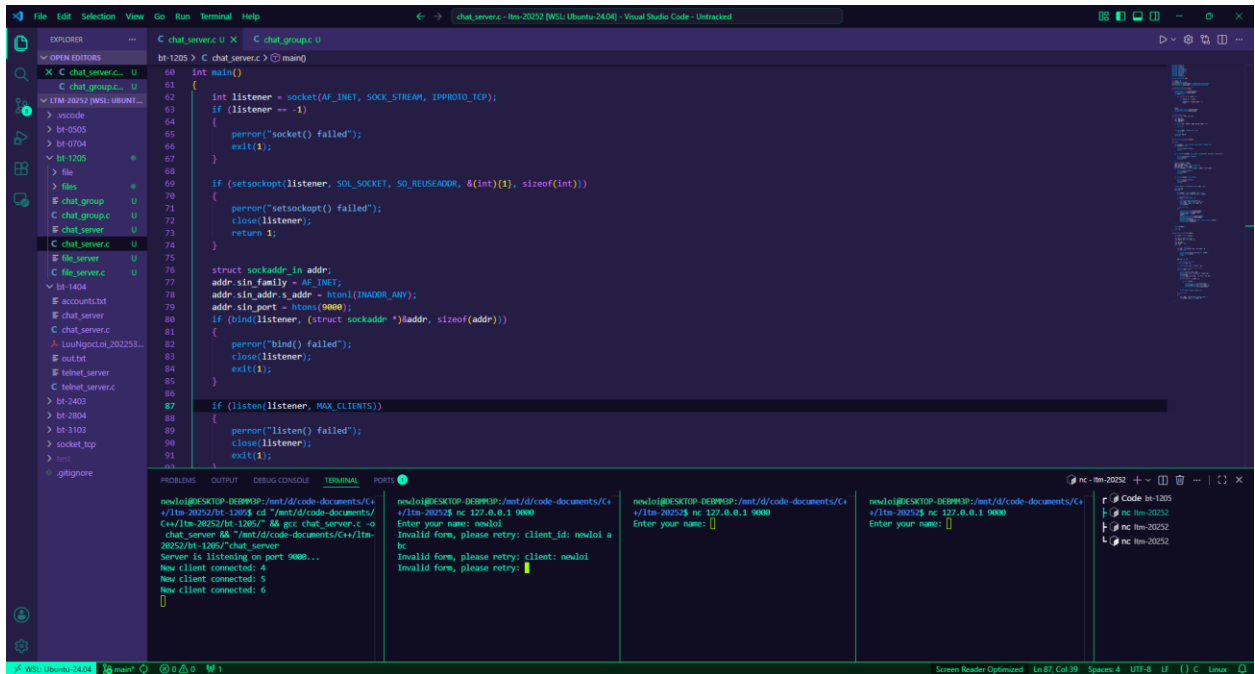
- Group gồm 2 client đầu tiên bắt đầu chat, client 3 không nhận được tin nhắn của group khác



- Chạy chat server và kết nối 3 client netcat



- Client nhập tên không đúng định dạng



- Các client nhập tên và gửi tới server

```
bt-1205 > C chat_server > main()
60 int main()
61 {
62     int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
63     if (listener == -1)
64     {
65         perror("socket() failed");
66         exit(1);
67     }
68
69     if (setsockopt(listener, SOL_SOCKET, SO_REUSEADDR, &(int){1}, sizeof(int)))
70     {
71         perror("setsockopt() failed");
72         close(listener);
73         return 1;
74     }
75
76     struct sockaddr_in addr;
77     addr.sin_family = AF_INET;
78     addr.sin_addr.s_addr = htonl(INADDR_ANY);
79     addr.sin_port = htons(9000);
80     if (bind(listener, (struct sockaddr *)&addr, sizeof(addr)))
81     {
82         perror("bind() failed");
83         close(listener);
84         exit(1);
85     }
86
87     if (listen(listener, MAX_CLIENTS))
88     {
89         perror("listen() failed");
90         close(listener);
91         exit(1);
92     }
93 }
```

newloi@DESKTOP-DEBP93P:/mnt/d/code-documents/C++/ltn-20252/bt-1205\$ cd "/mnt/d/code-documents/C++/ltn-20252/bt-1205/" && gcc chat_server.c -o chat_server && ./chat_server

Server is listening on port 9000...

New client connected: 4
New client connected: 5
New client connected: 6
Save client: id=newloi
Save client: id=loiloiloi
Save client: id=luangocloi

- Client 1 gửi tin nhắn

```
+ltn-20252/bt-1205$ cd "/mnt/d/code-documents/C++/ltn-20252/bt-1205/" && gcc chat_server.c -o chat_server && ./chat_server
```

Server is listening on port 9000...

New client connected: 4
New client connected: 5
New client connected: 6
Save client: id=newloi
Save client: id=loiloiloi
Save client: id=luangocloi
Receive: Hello moi nguoi
From client: 4 (newloi)
Sent to other clients.

newloi@DESKTOP-DEBP93P:/mnt/d/code-documents/C++/ltn-20252\$ nc 127.0.0.1 9000
Enter your name: newloi
Invalid form, please retry: client_id: newloi a bc
Invalid form, please retry: client: newloi
Invalid form, please retry: client_id: newloi
newloi: Hello moi nguoi

loiloiloi@DESKTOP-DEBP93P:/mnt/d/code-documents/C++/ltn-20252\$ nc 127.0.0.1 9000
Enter your name: client_id: loiloiloi
newloi: Hello moi nguoi

luangocloi@DESKTOP-DEBP93P:/mnt/d/code-documents/C++/ltn-20252\$ nc 127.0.0.1 9000
Enter your name: client_id: luangocloi
newloi: Hello moi nguoi

- Client 2 gửi tin nhắn

The screenshot displays a Windows 10 desktop environment. The primary focus is the Visual Studio Code (VS Code) editor, which is open to a C++ source file named `chat_group.c`. The code implements a multi-client chat server using sockets. It features a `main` function that sets up a listener socket on `AF_INET`, `SOCK_STREAM`, and `IPPROTO_TCP`. The server binds to `INADDR_ANY` on port `9000`. It then enters a loop where it accepts incoming connections, creates a new thread for each client, and returns control to the listener. The client threads handle sending and receiving messages from their respective sockets.

Below the editor, the `TERMINAL` pane is active, showing the execution of the program. The output indicates that the server is listening on port `9000` and successfully handles multiple client connections. The terminal output is as follows:

```
2025/12/15 12:05 "chat_server
Server is listening on port 9000...
New client connected: 4
New client connected: 5
New client connected: 6
Save client: id=ndoiol
Save client: id=loiolol
Save client: id=luangcol
Receive: Hello moi ngui
From client: 4 (ndoiol)
Sent to other clients.
Receives: Chao ban
From client: 5 (loiolol)
Sent to other clients.
[]
```

The terminal output is split across three windows, each showing the same sequence of events. The first window shows the server starting and the first client connecting. The second window shows the second client connecting and the first client sending a message. The third window shows the third client connecting and the second client sending a message.

Bài 4: telnet server

- Chạy telnet_server và 1 client netcat

- Client đăng nhập vào server (trường hợp thông tin tài khoản sai, hoặc gửi lệnh trước khi đăng nhập)


```
1 3
2 admin:admin
3 newlol:luongocloi
4 guest:nopass
```

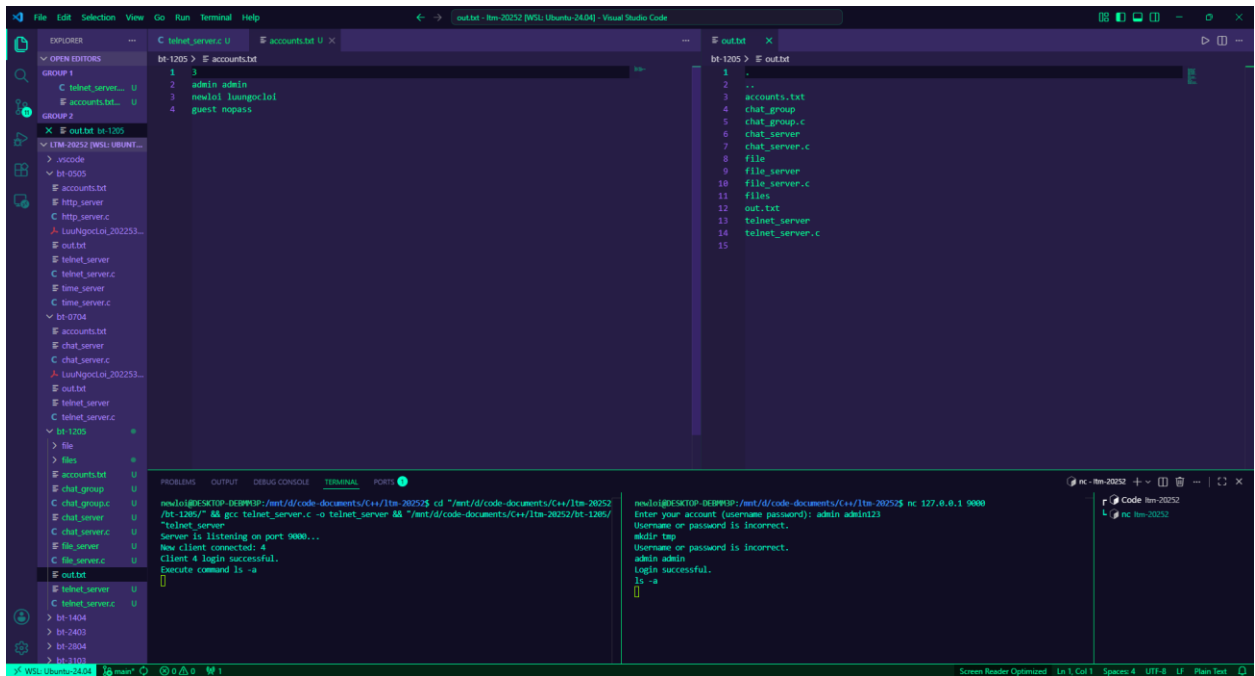
```
newlol@DESKTOP-DE3PP0P:/mnt/d/code-documents/C++/ltn-20252$ nc 127.0.0.1 9000
Enter your account (username password): admin admin123
Username or password is incorrect.
while true; do
    Username or password is incorrect.
done
```

- Client đăng nhập vào server (trường hợp thông tin tài khoản đúng)

```
newlol@DESKTOP-DE3PP0P:/mnt/d/code-documents/C++/ltn-20252$ cd "/mnt/d/code-documents/C++/ltn-20252/"
/hlt-1205/" && gcc telnet_server.c -o telnet_server && "/mnt/d/code-documents/C++/ltn-20252/hlt-1205/"
telnet_server
Server is listening on port 9000...
New client connected: 4
Client 4 login successful.
```

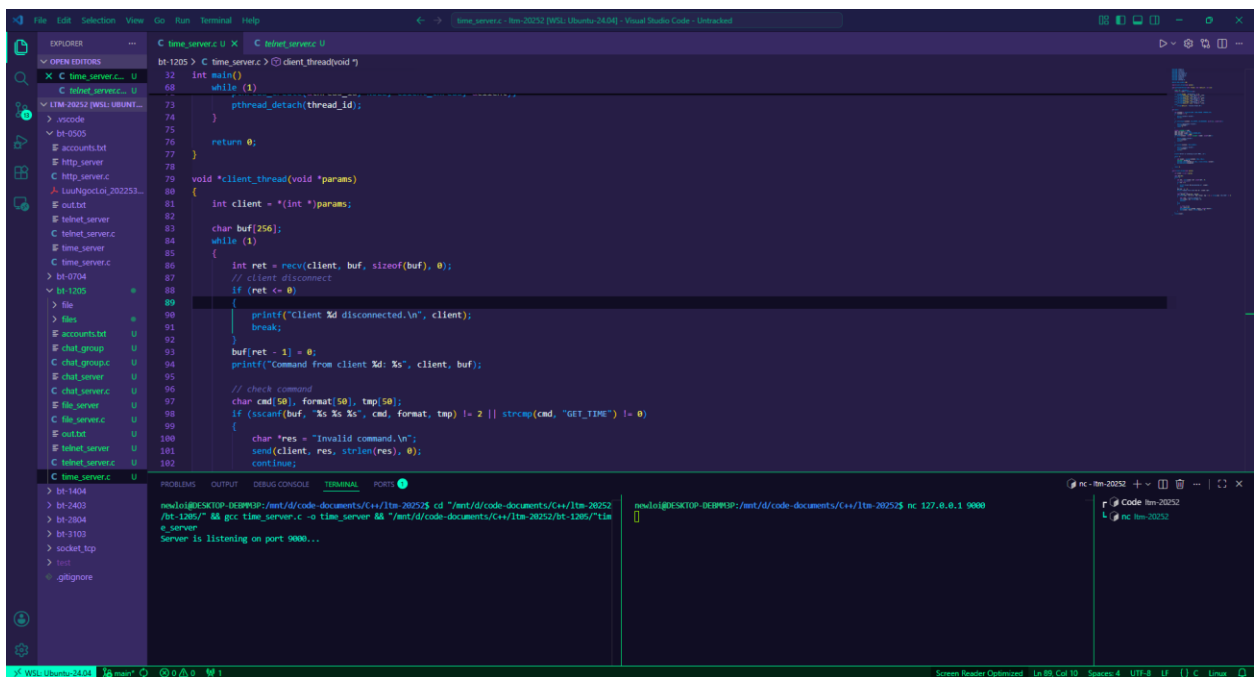
```
newlol@DESKTOP-DE3PP0P:/mnt/d/code-documents/C++/ltn-20252$ nc 127.0.0.1 9000
Enter your account (username password): admin admin
Username or password is incorrect.
while true; do
    Username or password is incorrect.
done
```

- Client gửi lệnh để server thực hiện (kết quả thực hiện lệnh ghi ra file out.txt)



Bài 5: time_server

- Chạy time_server và client netcat



- Client gửi lệnh tới server
 - o Sai lệnh

```
bt-1205 > C time_server.c > client_thread(void *)
32 int main()
68 while (1)
73 pthread_detach(thread_id);
74 }
75 }
76 return 0;
77 }
78 }
79 void *client_thread(void *params)
80 {
81 int client = *(int *)params;
82 char buf[256];
83 while (1)
84 {
85 int ret = recv(client, buf, sizeof(buf), 0);
86 // client disconnect
87 if (ret <= 0)
88 {
89 printf("Client %d disconnected.\n", client);
90 break;
91 }
92 buf[ret - 1] = 0;
93 printf("Command from client %d: %s", client, buf);
94 // check command
95 char cmd[50], format[50], tap[50];
96 if (sscanf(buf, "%s %s %s", cmd, format, tap) != 2 || strcmp(cmd, "GET_TIME") != 0)
97 {
98 char *res = "Invalid command.\n";
99 send(client, res, strlen(res), 0);
100 continue;
101 }
102 }
```

```
nc -l -p 9000
Server is listening on port 9000...
nc 127.0.0.1 9000
GET_TIME d/m/yyyy
Invalid command.
```

○ Sai định dạng thời gian

```
bt-1205 > C time_server.c > client_thread(void *)
32 int main()
68 while (1)
73 pthread_detach(thread_id);
74 }
75 }
76 return 0;
77 }
78 }
79 void *client_thread(void *params)
80 {
81 int client = *(int *)params;
82 char buf[256];
83 while (1)
84 {
85 int ret = recv(client, buf, sizeof(buf), 0);
86 // client disconnect
87 if (ret <= 0)
88 {
89 printf("Client %d disconnected.\n", client);
90 break;
91 }
92 buf[ret - 1] = 0;
93 printf("Command from client %d: %s", client, buf);
94 // check command
95 char cmd[50], format[50], tap[50];
96 if (sscanf(buf, "%s %s %s", cmd, format, tap) != 2 || strcmp(cmd, "GET_TIME") != 0)
97 {
98 char *res = "Invalid command.\n";
99 send(client, res, strlen(res), 0);
100 continue;
101 }
102 }
```

```
nc -l -p 9000
Server is listening on port 9000...
nc 127.0.0.1 9000
GET_TIME d/m/yyyy
Invalid command.
GET_TIME d/m/yyyy
Invalid format.
```

○ Đùng lệnh và định dạng

```

1205 > C time_server.c > client_thread(void *)
12 int main()
13 {
14     while (1)
15     {
16         pthread_detach(thread_id);
17     }
18     return 0;
19 }
20 void *client_thread(void *params)
21 {
22     int client = *(int *)params;
23     char buf[256];
24     while (1)
25     {
26         int ret = recv(client, buf, sizeof(buf), 0);
27         // client disconnect
28         if (ret <= 0)
29         {
30             printf("Client %d disconnected.\n", client);
31             break;
32         }
33         buf[ret - 1] = 0;
34         printf("Command from client %d: %s", client, buf);
35         // check command
36         char cmd[50], format[50], tmp[50];
37         if (sscanf(buf, "%s %s %s", cmd, format, tmp) != 2 || strcmp(cmd, "GET_TIME") != 0)
38         {
39             char *res = "Invalid command.\n";
40             send(client, res, strlen(res), 0);
41             continue;
42         }
43     }
44 }

```

Terminal output:

```

nc -Rm-20252 nc 127.0.0.1 9000
GET_TIME dm/mm/yy
24/05/2024
nc -Rm-20252 nc 127.0.0.1 9000
GET_TIME dm/mm/yy
24/05/26
Invalid command.
GET_TIME mm/dl/yy
05/24/2024

```

Bài 6: http_server

- Chạy http_server và 2 client netcat kết nối tới server sẽ nhận được response\
 - o Client 1

```

15 int main()
16 {
17     if (bind(listener, (struct sockaddr *)&addr, sizeof(addr)))
18     {
19         perror("listen() failed");
20         close(listener);
21         exit(1);
22     }
23     printf("Server is listening on port 9000...\n");
24     while (1)
25     {
26         int client = accept(listener, NULL, NULL);
27         pthread_t thread_id;
28         pthread_create(&thread_id, NULL, client_thread, &client);
29         pthread_detach(thread_id);
30     }
31     close(listener);
32     return 0;
33 }
34 void *client_thread(void *params)
35 {
36     int client = *(int *)params;
37     printf("New request from client: %d\n", client);
38     // return response to client
39 }

```

Terminal output:

```

nc -Rm-20252 nc 127.0.0.1 9000
HTTP/1.1 200 OK
Content-Type: text/html
<html><body>ch134Hello world</body></html>

```

- o Client 2

